Spectral estimates from black-box results II

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Elliptic lift

Given $\Omega \subset \mathbb{R}^d$ a smooth bounded open set. By the spectral theorem, we have

$$-\Delta e_k = \lambda_k^2 e_k,$$

where e_k is an orthonormal basis of $L^2(\Omega)$ and $\lambda_k^2 \in \mathbb{R}^+$. For $\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k$,

$$u(s,x) = \sum_{\lambda_k \leq \Lambda} \frac{\sinh(s\lambda_k)}{\lambda_k} u_k e_k(x), \qquad \begin{cases} (-\partial_s^2 - \Delta)u = 0, & \mathbb{R}_s \times \Omega \\ \partial_s u = \phi, & \Omega. \end{cases}$$

This shows that a spectral inequality

$$\|\phi\|_{L^2(\Omega)} \leq Ce^{c\Lambda} \|\phi\|_{L^2(\omega)},$$

is an unique continuation statement for u from $\{0\} \times \omega$.

Black-Box I: Carleman estimates

Theorem (JL99, Theorem 5.1)

Let $\omega \subset \Omega$ be a non-empty open set. There exists $K > 0$ such that for all $\Lambda > 0$ we have

$$\|\phi\|_{L^2(\Omega)} \leq e^{K\Lambda}\|\phi\|_{L^2(\omega)}, \qquad \phi \in \text{span}\{\lambda_k \mid \ \lambda_k \leq \Lambda\}.$$

Idea: Let $S_0 > 0$, $\beta \in (0, S_0/2)$. Let $Z := (0, S_0) \times \Omega$ and set $Y = (\beta, S_0 - \beta) \times \Omega$. We set $z = (s, x)$ with $s \in (0, S_0)$ and $x \in \Omega$.

$$Q := D_s^2 + P_0 + R, \quad \text{in } Z, \qquad Z \in \text{Diff}^1(Z)$$

Theorem (LRLR22, Theorem 7.12)

Let $\omega \subset \Omega$ open. Then $\exists C > 0 \delta \in (0,1)$ s.t $\forall u \in H^2(Z) : u\,|_{(0,S_0)\times\partial\Omega} = 0$ for $s \in (0, S_0)$,

$$\|u\|_{H^1(Y)} \leq C\|u\|^{1-\delta}_{H^1(Z)} \left(\|Qu\|_{L^2(Z)} + |u\,|_{s=0}|_{H^1(\omega)} + |\partial_s u\,|_{s=0}|_{L^2(\omega)}\right)^{\delta}$$

Black-Box II: Propagation of smallness

Recall the $n-$Hausdorff measure of a set $E \subset \mathbb{R}^d$,

$$\mathcal{C}^n_{\mathcal{H}}(E) = \text{ínf} \left\{ \sum_j r_j^n \mid \ E \subset \cup_j B(x_j, r_j) \right\},$$

and the Hausdorff dimension of E defined as

$$\dim_{\mathcal{H}}(E) = \text{ínf}\{n \mid \ \mathcal{C}^d_{\mathcal{H}}(E) = 0\}.$$

Denote by $|E|$ the Lebesgue measure of E. The $d-$Hausdorff measure is equivalent to the $d-$Lebesgue measure: there exists $C_d, c_d > 0$ such that for every Borel set

$$c_d|A| \leq \mathcal{C}^d_{\mathcal{H}}(A) \leq C_d|A|,$$

and

$$\mathcal{C}^d_{\mathcal{H}}(E) > 0 \implies \forall d' < d, \ \mathcal{C}\mathcal{C}^{d'}_{\mathcal{H}}(E) \geq \text{ínf}\{1, \mathcal{C}^d_{\mathcal{H}}(E)\}.$$

Observation

The value of the Hausdorff measure is not invariant by diffeomorphisms, but it is by Lipschitz diffeomorphisms.

Let $(\mathcal{M}, g)$ be a $W^{2,\infty}$ compact Riemannian manifold with boundary and let Δ be the Laplace operator on M:

$$\Delta = \frac{1}{\kappa(x)} \operatorname{div}(g^{-1}(x)\kappa(x)\Delta_x) = \operatorname{div}_\kappa \nabla_g,$$

where κ is a positive Lipschitz density. For instance, we recover the usual Laplace-Beltrami operator when $k(x) = |g|^{1/2}$.

$$\begin{cases} -\Delta e_k = \lambda_k^2 e_k, & M, \\ \alpha e_k + \beta \partial_\nu e_k = 0, & \partial M, \end{cases}$$

where $\alpha, \beta \in \{0, 1\}$ and $\alpha + \beta = 1$.

Theorem (BM23, Theorem 1)

There exists $\delta \in (0,1)$, depending only on the dimension of the manifold, such that for any $m > 0$, there exists $C, D > 0$ such that for any $E_1 \subset M$ with $|E_1| \geq m$ and any $E_2 \subset M$ sastisfying

$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) > m, \tag{1}$$

and any $\Lambda > 0$, we have

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \implies \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \|\phi \mathbf{1}_{E_1}\|_{L^1(M)}, \tag{2}$$

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \implies \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \sup_{x \in E_2} |\phi \mathbf{1}_{E_1}|, \tag{3}$$

Observation

Since the only requirement on E_2 is $|E_2| > 0$, the result applies even when E_2 is concentrated arbitrarily close to ∂M.

New three spheres theorem

Let u be a solution of an elliptic equation in divergence form in a bounded domain $\Omega \subset \mathbb{R}^n$,

$$\operatorname{div}(A\nabla u) = 0,$$

where $A(x) = [a_{ij}(x)]_{1\leq i,j\leq n}$ is a symmetric uniformly elliptic matrix with Lipschitz entries,

$$\Lambda_1^{-1}\|\zeta\|^2 \leq \langle A\zeta, \zeta\rangle \leq \Lambda_1\|\zeta\|^2, \quad |a_{ij}(x) - a_{ij}(y)| \leq \Lambda_2|x-y|,$$

where $\Lambda_1 \geq 1$ and $\Lambda_2 \geq 0$.

Logunov-Malinnikova's three spheres inequality

Theorem ([?], Theorem 5.1)

There exists a constant $c \in (0,1)$ that depends only on the dimension n such that the following holds. Let $m, \delta, \rho > 0$ and suppose that $E, \mathcal{K} \subset \Omega$ satisfies

$$\mathcal{C}_{\mathcal{H}}^{n-1-c+\delta}(E) > m, \quad \operatorname{dist}(E, \partial\Omega) > \rho, \quad \operatorname{dist}(\mathcal{K}, \partial\Omega) > \rho.$$

Then there exist $C, \gamma > 0$ depending on $m, \delta, \rho, \Lambda_1, \Lambda_2, \Omega$ such that

$$\sup_{\mathcal{K}} |\nabla u| \leq C \left(\sup_{E} |\nabla u| \right)^{\gamma} \left(\sup_{\Omega} |\nabla u| \right)^{1-\gamma}.$$

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About the program

Today's plan

1. Prove Burq-Moyano's Theorem 1. [BM23]
2. See some control-observability consequences (without seeing the double manifold)
3. Extend previous results to Burq-Moyano's work on heat equation on $\mathbb{R}^d$ [BM25].

Next week:

1. The double manifold
2. Simultaneous Dirichlet and Neumann heat equations [Burq-Moyano]
3. Quantitative smallness prop. for Schrödinger op. [Le Balc'h-Martin]
4. Maybe: On the dimension of observable sets for the heat equation [Walton Green, Le Balc'h, Martin, Orsoni]
   Multiplication of spec. ineq??

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Proof pt. 1: Patch estimate

Last week: (3) implies (2). Now we check the boundaryless case.

Idea: Reduce the problem to LM's three spheres inequality.

Let $m > 0$ and $E_2 \subset M$: $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) > m$. By compactness $M \subset \cup_{j=1}^{N} U_j$ with diffeos. $\psi_j : V_j \to \mathbb{R}^d$ with $U_j \subset V_j$. By Jerison-Lebeau's, there exists $C, D > 0$ depending only on M such that for any $j \in \{1, \dots, N\}$,

$$\|\phi\|_{L^2(M)} \leq Ce^{d\Lambda}\|\phi\|_{L^2(U_j)}.$$

Let j_0 be such that

$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2 \cap U_{j_0}) > \frac{m}{N}.$$

$$V := \psi_{j_0}(V_{j_0}), \quad U := \psi_{j_0}(U_{j_0}), \quad F := \psi_{j_0}(E_2 \cap U_{j_0}).$$

Since ψ_{j_0} is a $W^{2,\infty}$ diffeo,

$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(F) > C(\psi_0)\frac{m}{N}.$$

Proof pt. 2: Getting the elliptic eq.

Write $f_k = (\psi_{j_0})_* e_k, \varphi = (\psi_{j_0})_* \phi$. We get

$$\frac{1}{\kappa(x)} \operatorname{div}_x \left(\kappa(x) g^{-1}(x) \nabla_x f_k\right) = -\lambda_k^2 f_k, \quad \text{on } V.$$

Set

$$u(t,x) = \sum_{\lambda_k \leq \Lambda} \frac{\sinh(\lambda_k t)}{\lambda_k} u_k f_k(x), \quad \varphi(x) := \sum_{\lambda_k \leq \Lambda} u_k f_k(x) = \partial_t u \,|_{t=0} .$$

Then we have the elliptic eq.

$$\left(\operatorname{div}(g^{-1}\kappa \nabla_x) + \partial_t \kappa \partial_t\right) u = 0.$$

For $T_2 > T_1 > 0$, let us define

$$E = \{0\} \times F, \quad K := [-T_1, T_1] \times \overline{U}, \quad \Omega = (-T_2, T_2) \times V,$$

which by construction satisfy the inclusions $E \subset K \subset \Omega$.

Proof pt. 3: LM18 estimate

Using (3) with $d = n-1$ and $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E) = \mathcal{C}_{\mathcal{H}}^{d-\delta}(F)$,

$$\mathcal{C}_{\mathcal{H}}^{n-1-\delta}(E) > m', \quad n = d+1, \; m' = C(\psi_0)\frac{m}{N}.$$

For $\delta > 0$ small, we have from [LM18, Theorem 5.1],

$$\sup_K |\nabla_{t,x} u| \leq C \left(\sup_E |\nabla_{t,x} u| \right)^{\alpha} \left(\sup_\Omega |\nabla_{t,x} u| \right)^{1-\alpha}. \tag{4}$$

Lemma

[?] There exists $\sigma > 0$ such that

$$\|\nabla_x u\|_{L^\infty} + \|u\|_{L^\infty} \leq C\|u\|_{\mathcal{H}^\sigma},$$

with $\mathcal{H}^\sigma := D((-\Delta)^{\sigma/2})$ endowed with $\|\phi\|_{\mathcal{H}^\sigma}^2 = \sum_k (1+\lambda_k)^{2\sigma}|u_k|^2$.

Properly concatenating of the previous estimates yields the result.

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Theorem (BM23, Theorem 2.)

Let $F \subset (0,T) \times M$ of positive Lebesgue measure. Then there exists $C > 0$ such that for any $u_0 \in L^2(M)$ the solution $u = e^{t\Delta} u_0$ to the heat equation

$$(\partial_t - \Delta) u = 0, \quad u \mid_{\partial M} = 0 \text{ or } \partial_\nu u \mid_{\partial M} = 0, \quad u(t,0) = u_0, .$$

satisfies

$$\| e^{T\Delta} u_0 \|_{L^2(M)} \leq C \int_F |u(t,x)| \kappa(x) \, dxdt.$$

As a consequence, for all $u_0, v_0 \in L^2(M)$, there exists $f \in L^\infty(F)$ such that the solution to

$$(\partial_t - \Delta) u = f \mathbf{1}_F(t,x), \quad u \mid_{\partial M} = 0 \text{ or } \partial_\nu u \mid_{\partial M} = 0, \quad u(t,0) = u_0,$$

satisfies $u \mid_{t=T} = e^{T\Delta} v_0$.

Theorem (BM23, Theorem 3.)

There exists $\delta \in (0,1)$ depending on the dimension of M such that for any $E \subset M$ with positive $(d-\delta)$-Hausdorff content and any $J \subset (0,T)$ of positive Lebesgue measure, there exists $C > 0$ such that for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to

$$(\partial_t - \Delta)u = 0, \quad u\,|_{\partial M} = 0 \text{ or } \partial_\nu u\,|_{\partial M} = 0,$$

satisfies

$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \int_J \sup_{x\in E} |u(t,x)|dt.$$

As a consequence, assuming additionally that E is closed in M, for all $u_0, v_0 \in L^2(M)$ there exists a Borel measure μ supported on $(0,T)\times E$ such that the solution to

$$(\partial_t - \Delta)u = \mu(t,x)\mathbf{1}_{J\times E}, \quad u\,|_{\partial M} = 0 \text{ or } \partial_\nu u\,|_{\partial M} = 0, \quad u(0,x) = u_0,$$

satisfies $u\,|_{t\geq T} = e^{t\Delta}v_0$.

Theorem (BM23, Theorem 4)

Take $\delta \in (0,1)$ as in Theorem 1 [?]. Let $m > 0$, $\tau \in (0,1)$ and $D > 0$. There exists $C > 0$ such that if $E_1 \subset M$ satisfies $|E_1| \geq m$ or $E_2 \subset M$ satisfies $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) \geq m$. Then for any sequence $(s_n)_{n\in\mathbb{N}}$,

$$0 < \cdots < s_n < \cdots < s_0 < T,$$

converging not too fast to 0, i.e,

$$\exists \tau \in (0,1),\ \forall n \in \mathbb{N},\ (s_n - s_{n+1}) \geq \tau(s_{n-1} - s_n),$$

we have for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the heat equation,

$$(\partial_t - \Delta)u = f\mathbf{1}_F(t,x), \quad u\,|_{\partial M} = 0 \text{ or } \partial_\nu u\,|_{\partial M} = 0, \quad u(t,0) = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \sup_{n\in\mathbb{N}} e^{-\frac{D}{s_n - s_{n+1}}} \int_{E_1} |e^{s_n\Delta}u_0|(s_n,x)|dx \tag{5}$$

Theorem (BM23, Theorem 4)

and

$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \sup_{n\in\mathbb{N},x\in E_2} e^{-\frac{D}{s_n-s_{n+1}}}|e^{s_n\Delta}u_0|(s_n,x)|. \tag{6}$$

As a consequence, if E_1 is a closed subset of M, given any $(t_n)_{n\in\mathbb{N}}$ with $0 < t_0 < \cdots < t_n < \cdots < T$ converging not too fast to T. For all $u_0, v_0 \in L^2(M)$, there exists (f_j) a sequence of functions on E_1 such that

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}}\|f\|_{L^\infty(E_1)} < +\infty,$$

and the solution to

$$\begin{cases} (\partial_t - \Delta)u = \sum_{j=1}^{\infty}\delta_{t=t_j}\otimes f_j(x)\mathbf{1}_{E_1},, & \mathbb{R}_+\times M, \\ u = 0, \quad or \quad \partial_\nu u = 0, & \mathbb{R}_+\times\partial M, \\ u(0,x) = u_0, & M, \end{cases} \tag{7}$$

satisfies $u\,|_{t>T} = e^{t\Delta}v_0$.

Theorem (BM23, Theorem 4)

Similarly, under the additional assumption that E_2 is a closed subset of M, given any sequence $(t_n)_{n\in\mathbb{N}}$ with $0 \leq t_0 < \cdots < t_n < \cdots < T$ converging not too fast to T, for all $u_0, v_0 \in L^2(M)$, there exists (μ_j) a sequence of Borel measure supported on E_2 such that

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}} |\mu_j|(E_2) < +\infty,$$

and the solution to

$$\begin{cases} (\partial_t - \Delta)u = \sum_{j=1}^{\infty} \delta_{t=t_j} \otimes \mu_j(x)\mathbf{1}_{E_2}, & \mathbb{R}_+ \times M, \\ u = 0, \quad or \quad \partial_\nu u = 0, & \mathbb{R}_+ \times \partial M, \\ u(0,x) = u_0, & M, \end{cases} \tag{8}$$

satisfies $u =|_{t>T}= e^{t\Delta}v_0$.

Some comments on the proof

1. We need the double manifold: $\tilde{M} = \overline{M} \times \{-1, 1\}/\partial M$. Topologically, $\tilde{M}$ comes from gluing M_D and M_N through the boundary.
2. Theorem 2 is proven following direct Lebeau-Robbiano's black box from Miller's work!.
3. Theorem 3 and 4 comes from double manifold + Hahn-Banach Theorem + Riesz Representation Theorem + Duality arguments.

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Burq-Moyano 25': Setting

We consider on $\mathbb{R}^d$ the following Laplace operator

$$\Delta = \frac{1}{\kappa(x)} \sum_{i,j} \partial_{x_i} g^{ij}(x) \kappa(x) \partial_{x_j},$$

where we assume that the coefficients κ, g are Lipschitz,

$$\|\kappa\|_{W^{1,\infty}(\mathbb{R}^d)} + \|g\|_{W^{1,\infty}(\mathbb{R}^d)} \leq A, \tag{9}$$

and that g is uniformly elliptic,

$$\exists a > 0 : \forall x \in \mathbb{R}^d, \forall X \in \mathbb{R}^d, \quad a\|X\|^2 \leq (g(x)X, X), \qquad \text{and} \quad \kappa(x) \geq a. \tag{10}$$

Let $\omega \subset \mathbb{R}^d$ satisfying the following condition

$$\exists R, \delta > 0, \quad \forall x \in \mathbb{R}^d, \ \text{meas}(\omega \cap B(x, R)) \geq \delta, \tag{11}$$

or

$$\exists R, \delta > 0, \quad \forall x \in \mathbb{R}^d, \ \mathcal{C}^n_{\mathcal{H}}(\omega \cap B(x, R)) \geq \delta. \tag{12}$$

Burq-Moyano 25': Setting

Let us recall that $(-\Delta, H^2(\mathbb{R}^d))$ on $L^2(\mathbb{R}^d, \kappa(x)dx)$ is self-adjoint with non-negative continuous spectrum. By SDT,

$$u = \int_0^{+\infty} dm_\lambda u, \qquad \Pi_\mu(u) = \int_0^{\mu} dm_\lambda u, \qquad \forall u \in L^2(\mathbb{R}^d, \kappa(x)dx),$$

where $\mu > 0$ and dm_λ denotes the spectral measure of the operator $\sqrt{-\Delta}$. Moreover, if ϕ, φ are bounded functions on $\mathbb{R}$ with real values, we have

$$\phi(\sqrt{-\Delta})u = \int_0^{+\infty} \phi(\lambda) dm_\lambda u,$$

$$(\phi(\sqrt{-\Delta})u, \varphi(\sqrt{-\Delta})u)_{L^2(\mathbb{R}^d, \kappa(x)dx)} = \int_0^{+\infty} \phi(\lambda)\varphi(\lambda)(dm_\lambda u, u).$$

and consequently

$$\|\phi(\sqrt{-\Delta})\Pi_\mu u\|_{L^2(\kappa dx)} \leq \sup_{\lambda \in [0,\mu]} |\phi(\lambda)| \|u\|_{L^2(\kappa dx)}.$$

Theorem (BM25, Theorem 1)

Assume that ω satisfies assumption (11). Then there exists $C > 0$ such that for any $u \in L^2(\mathbb{R}^d)$,

$$\|\Pi_\mu u\|_{L^2(\mathbb{R}^d;\kappa dx)} \leq Ce^{C\mu}\|\Pi_\mu u\|_{L^2(\omega;\kappa dx)}. \tag{13}$$

Assume now that ω verifies assumption (12) for some $n < d$ sufficiently close to d. Then there exists $C > 0$ such that for any $u \in L^2(\mathbb{R}^d)$,

$$\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d;\kappa dx)} \leq Ce^{C\mu}\sum_{k\in\mathbb{Z}}\|\Pi_\mu u\|^2_{L^\infty(\omega\cap B(k,R))}. \tag{14}$$

Theorem (BM25,Theorem 2.)

Let $F \subset (0,T)$ be a set of positive measure. Let $E \subset \mathbb{R}^d$ satisfying (11). Then there exists $C > 0$ such that for any $u_0 \in L^2(\mathbb{R}^d)$, the solution $u = e^{t\Delta}u_0$ to the heat equation

$$\partial_t u - \Delta u = 0, \qquad u\,|_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(\mathbb{R}^d)} \leq C \int_{F\times E} |u(t,x)|^2 \kappa(x)\,dxdt. \tag{15}$$

As a consequence, for all $u_0, v_0 \in L^2(\mathbb{R}^d)$ there exists $f \in L^2(F\times E)$ such that the solution to

$$(\partial_t - \Delta)u = f\mathbf{1}_{F\times E}(t,x), \quad u\,|_{t=0} = u_0,$$

satisfies

$$u\,|_{t=T} = e^{T\Delta}v_0.$$

Theorem (BM25, Theorem 3.)

Let $F \subset (0,T)$ be a set of positive measure. Let $E \subset \mathbb{R}^d$ satisfying (12). Then, there exists $C > 0$ such that for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the heat equation

$$\partial_t u - \Delta u = 0, \quad u\,|_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(\mathbb{R}^d)} \leq C \sum_{k\in\mathbb{Z}^d} \int_F \sup_{x\in E\cap B(k,R)} |u(t,x)|^2 dt. \tag{16}$$

Theorem (BM25, Theorem 4.)

Let $m > 0, \tau \in (0,1)$ and $D > 0$. There exists $C > 0$, such that if $E \subset \mathbb{R}^d$ satisfies (11), then for any sequence $(s_n)_{n\in\mathbb{N}}$,

$$J = \{0 < \cdots < s_n < \cdots < s_0 < T\},$$

converging not too fast to 0, i.e,

$$\exists \tau \in (0,1), \forall n \in \mathbb{N}, (s_n - s_{n+1}) \geq \tau(s_{n-1} - s_n),$$

we have that for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the eat equation

$$\partial_t u - \Delta u = 0, \quad u\,|_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(M)} \leq C \sup_{n\in\mathbb{N}} e^{-\frac{D}{s_n - s_{n+1}}} \int_E |e^{s_n\Delta}u_0|^2(s_n, x)dx. \tag{17}$$

Theorem (BM25, Theorem 4)

As a consequence, given any sequence $(t_n)_{n\in\mathbb{N}}$,

$$J = \{0 < t_0 < \cdots < t_n < \cdots < T\},$$

converging not too fast to T. *For all* $u_0, v_0 \in L^2(M)$ *there exists* $(f_j)_j$ *a sequence of functions on* E_1 *s.t*

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}} \|f\|_{L^2(E)} < +\infty,$$

and the solution to

$$(\partial_t - \Delta)u = \sum_{j=1}^{\infty} \delta_{t=t_j} \otimes f_j(x)\mathbf{1}_{E_1}, \tag{18}$$

satisfies $u\,|_{t=T} = e^{t\Delta}v_0$.

Proof Theorem 1.

$$v_\mu(t,x) = \int_{\lambda=0}^{\mu} \frac{\sinh(\lambda t)}{\lambda} dm_\lambda u. \tag{19}$$

Recall that

$$-\Delta v_\mu = \int_{\lambda=0}^{\mu} \frac{\sinh(\lambda t)}{\lambda} \lambda^2 dm_\lambda u.$$

As a consequence, we have

$$\left(\frac{1}{\kappa} \operatorname{div} g^{-1} \kappa \nabla_x + \partial_t^2\right) v_\mu = 0 \Longrightarrow (\operatorname{div} g^{-1} \kappa \nabla_x + \partial_t \kappa \partial_t) v_\mu = 0.$$

Let $1 = \sum_p \chi(x-p)$ be a partition of unity associated. Consider, for $T_2 > T_1 > 0$ the sets

$$K_p := [-T_1, T_1] \times \overline{B(p,R)}, \qquad \Omega_p := (-T_2, T_2) \times B(p, 2R),$$
$$F_p := \omega \cap B(p,R), \qquad E_p := \{0\} \times F_p,$$

which by construction satisfy the inclusions $E_p \subset K_p \subset \Omega_p$.

Proof Theorem 1: First assumption

Now assumption (11) ensures that $|F_p| \geq \delta$. Note that by Hausdorff measure, since F_p has a diameter at most $2R$, when computing the Hausdorff content of F_p, we can assume that for every j, we have $r_j \leq 4R$. As a consequence,

$$\mathcal{C}^{d-1-\varepsilon}(E_p) \geq (4R)^{-\varepsilon}\mathcal{C}^{d-1}(E_p) \geq c\delta.$$

By LM18,

$$\sup_{K_p} |\nabla_{t,x} v_\mu| \leq C \left(\sup_{E_p} |\nabla_{t,x} v_\mu| \right)^{\alpha} \left(\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha} \tag{20}$$

Proof Theorem 1: First assumption

we replace L^∞ norm by L^2 norm. Let

$$a = \left(\varepsilon \frac{\sup_{K_p} |\nabla_{t,x} v_\mu}{\sup_{\Omega_p} |\nabla_{t,x} v_\mu|} \right)^{1/a}, \quad E_p' = \{x \in E_p : |\nabla_{t,x} v_\mu| \leq a\}.$$

By LM18, for $\varepsilon C < 1$, $|E_p'| \leq \delta/2$. As a consequence,

$$\int_{E_p} |\nabla_{t,x} v_\mu|^2 dx \geq \int_{E \setminus E_p'} |\nabla_{t,x} v_\mu|^2 dx \geq a^2 \frac{\delta}{2} \geq \frac{\delta}{2} \left(\varepsilon \frac{\sup_{K_p} |\nabla_{t,x} v_\mu|}{(\sup_{\Omega_p} |\nabla_{t,x} v_\mu|)^{1-\alpha}} \right)^{2/\alpha}$$

and we deduce

$$\|\nabla_{t,x} v_\mu\|_{L^2(B(p,R))} \leq C \left(|\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha} \|\nabla_{t,x} v_\mu\|^{\alpha}_{L^2(E \cap B(p,R))}. \quad (21)$$

Proof Theorem 1: First assumption

Lemma ([?], Lemma 2.1)

There exists $C, K > 0$ such that for all $p \in \mathbb{Z}^d$, and all $u \in L^2(\mathbb{R}^d; \kappa(x)dx)$,

$$\left(\sum_{p \in \mathbb{Z}^d} \|\nabla_{t,x} v_\mu\|^2_{L^\infty((-T_2,T_2)\times B(p,2R))} \right)^{1/2} \leq Ce^{K\mu} \|\Pi_\mu u\|_{L^2(\mathbb{R}^d;\kappa(x)dx)}.$$

By Young's ineq, $a^{2-2\alpha} b^{2\alpha} \leq C(a^2 + b^2)$, hence previous estimate yields

$$\|\Pi_\mu u\|^2_{L^2(B(p,R))} \leq Ce^{-D\mu} \sup_{\Omega_p} |\nabla_{t,x} v_\mu|^2 + Ce^{\frac{2-2\alpha}{2\alpha} D\mu} \|\Pi_\mu u\|^2_{L^2(E\cap B(p,R))}.$$

Now, summing over $p \in \mathbb{Z}$ and using the lemma, we get the result for μ large enough taking $D > 2K$ since

$$\|\Pi_\mu u\|^2_{L^2(B(p,R))} \leq Ce^{-D\mu} e^{2K\mu} \|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} + Ce^{D\frac{2-2\alpha}{2\alpha}\lambda} \|\Pi_\mu u\|^2_{L^2(E)},$$

Proof Theorem 1: Second assumption

Suppose $\mathcal{C}^n(E_p) \geq \delta$, with n is sufficiently close to d. By LM18,

$$\sup_{K_p} |\nabla_{t,x} v_\mu| \leq C \left(\sup_{E_p} |\nabla_{t,x} v_\mu| \right)^{\alpha} \left(\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha},$$

where the constant is uniform with respect to p because the metric and κ are bounded on $B(p, 2R)$, since the metric is uniformly elliptic. We deduce from $\Pi_\mu u = \partial_t v_\mu \mid_{t=0}$,

$$\|\Pi_\mu u\|_{L^2(B(p,R))} \leq C \sup_{K_p} |\nabla_{t,x} v_\mu| \leq C \left(\sup_{E_p} |\nabla_{t,x} v_\mu| \right)^{\alpha} \left(\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha},$$

and we deduce for all $D > 0$,

$$\|\Pi_\mu u\|^2_{L^2(B(p,R))} \leq C e^{-D\mu} \sup_{\Omega_p} |\nabla_{t,x} v|^2 + C e^{\frac{2-2\alpha}{2\alpha} D\mu} \|\Pi_\mu u\|^2_{L^\infty(E \cap B(p,R))}.$$

Proof Theorem 2: Second assumption

Now, from the Sobolev embedding we deduce

$$\begin{aligned}\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} &\leq Ce^{-D\mu}\sum_{p\in\mathbb{Z}}\sup_{\Omega_p}|\nabla_{t,x}v|^2 + Ce^{D\frac{2-2\alpha}{2\alpha}\mu}\sum_{p\in\mathbb{Z}}\|\Pi_\mu u\|^2_{L^\infty((E\cap B(p,R))\mathbb{R}^d}\\ &\leq Ce^{-D\mu}e^{2K\mu}\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} + Ce^{D\frac{2-2\alpha}{2\alpha}\lambda}\sum_{p\in\mathbb{Z}^d}\|\Pi_\mu u\|^2_{L^\infty(E\cap B(p,R))},\end{aligned}$$

and the proof of the second part for $\mu \geq \mu_0$ follows from taking $D > 2K$. Finally, for $\mu < \mu_0$, apply Theorem 1 for μ_0 to $u = \Pi_\mu(u)$. □